

# RASHEDUL ISLAM

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## PROFESSIONAL SUMMARY

Computer Science student and AI/ML researcher with hands-on experience building full-stack applications, machine learning systems, and cybersecurity tools. Proficient in Python, Java, JavaScript, and C/C++ with a strong foundation in data structures and algorithms. Experienced with microservices architecture, cloud deployment, and security-first development practices. Actively contributing to open-source projects and competitive programming platforms.

## EDUCATION

**B.Sc. in Computer Science and Engineering** Southeast University, Dhaka, Bangladesh Nov 2023 – 2027  
(Expected)

- CGPA: 3.9 out of 4.0
- Relevant Coursework: Data Structures & Algorithms, OOP, Database Management, Operating Systems, Computer Networks, Discrete Mathematics, Linear Algebra, Probability & Statistics
- Active competitive programmer — solved 200+ problems on Codeforces and LeetCode; consistently placing in top 30% of rated contests

## TECHNICAL SKILLS

|                           |                                                                                             |
|---------------------------|---------------------------------------------------------------------------------------------|
| <b>Languages</b>          | Python, Java, C, C++, JavaScript, PHP, SQL                                                  |
| <b>AI / ML</b>            | TensorFlow, PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, Transformers (BERT)  |
| <b>Web Frameworks</b>     | FastAPI, Spring Boot, Django, React                                                         |
| <b>Databases</b>          | MySQL, PostgreSQL, MongoDB, Oracle Database                                                 |
| <b>DevOps &amp; Cloud</b> | Docker, AWS (EC2, S3, Lambda), Git, GitHub Actions, Linux, Maven                            |
| <b>Cybersecurity</b>      | Penetration testing, OWASP Top 10, Burp Suite, threat modelling, privilege escalation, CTF  |
| <b>Blockchain</b>         | Decentralised app architecture, smart contract fundamentals, chain-of-trust protocol design |

## PROJECTS

**AgriLink — Agricultural Marketplace System** — *Java, JavaFX, MySQL* 2024

- Built a full-stack desktop marketplace connecting farmers, buyers, and suppliers with inventory management and order tracking
- Designed a relational database schema supporting 8+ entities; implemented role-based access for three distinct user types
- Delivered a responsive JavaFX UI with real-time data binding and ~50 ms average query response time

**API Tester & Request Logger** — *PHP, Vanilla JS, MySQL, XAMPP* 2024

- Developed a lightweight Postman-inspired browser tool to test REST API endpoints, log requests, and save favourites
- Built a persistent request history with MySQL backend, supporting filtering and replay of past API calls
- Achieved single-page app feel using vanilla JS async/fetch — no framework overhead, sub-100 ms UI interactions

**Real-Time Weather App** — *Python, Tkinter, OpenWeatherMap API* 2025

- Built a desktop GUI application that fetches and displays live atmospheric data (temperature, humidity, wind speed, pressure, cloud coverage) for any city worldwide using the OpenWeatherMap REST API
- Integrated HTTP request handling and JSON response parsing via the Requests library, with smart error handling for invalid city inputs
- Designed a clean Tkinter interface delivering instant weather lookups with support for 60 API calls/minute on the free tier

**Patient Management System** — *C, Data Structures* 2023

- Built a console-based healthcare record system in C implementing a queue (FIFO) for patient management and a stack (LIFO) for bill processing, demonstrating applied data structures in a real-world domain
- Developed core features including patient add/delete/search, bill management, due notifications, and payment receipt generation with automatic 25% discount logic for patients aged 50+
- Designed modular functions (addPatient, deletePatient, searchPatient, manageBills, paymentReceipt) with clean separation of concerns across a single-file C implementation

## RESEARCH & ACTIVITIES

### ML & Cybersecurity Researcher (Self-Directed)

2024 – Present

- Investigating deep learning for anomaly detection in network traffic and adversarial robustness of classification models
- Exploring the intersection of AI safety and network security in low-resource and adversarial environments

### Competitive Programmer — Codeforces / LeetCode

2024 – Present

- Solved 200+ problems spanning dynamic programming, graph algorithms, segment trees, and number theory
- Consistently placing in top 30% of rated contests; applying systematic problem decomposition and analytical thinking

### CTF Participant & Penetration Testing Practitioner — TryHackMe

2024 – Present

- Completed **160+** rooms covering offensive security labs, OWASP Top 10 vulnerabilities, privilege escalation, network enumeration, and web exploitation
- Ranked among active practitioners on TryHackMe; consistently completing hands-on labs across beginner to intermediate difficulty tracks
- Applied security-first mindset across all web projects: input validation, secure headers, secrets management

## CERTIFICATIONS

Intro to Machine Learning — Kaggle

Apr 2026

Introduction to Cybersecurity — Cisco Networking Academy

Apr 2026

Getting Started with DevOps on AWS — Amazon Web Services

Mar 2026

Exploring AI Use Cases and Applications — Amazon Web Services

Mar 2026

Certified Ethical Hacking Masterclass — MSB Academy

2025

Fundamentals of Machine Learning and Artificial Intelligence — AWS

Dec 2025

Intro to SQL — Kaggle

Feb 2026

Python — Kaggle

Feb 2026

Intro to Programming — Kaggle

Jan 2026

Fundamentals of Machine Learning and AI — IBM

2024

Exploring AI Use Cases and Applications — IBM

2024

Intro to Programming with Python — IBM

2024

Intro to SQL — IBM

2024

## ADDITIONAL INFORMATION

**Languages** Bengali (Native), English (Professional Proficiency)

**Open Source** Active contributor on GitHub; advocate for clean, documented, and tested code

**Interests** Distributed systems, large-scale ML infrastructure, developer tooling, AI safety & cybersecurity

**Availability** Open to remote internship and full-time opportunities globally; flexible start date